Formula Sheet

CBSE Class 12 Physics • NCERT Part II, Chapters 9–14

Blue boxes are NCERT (board) formulas, with SI units in [brackets]. Purple **JEE extra** boxes go beyond the NCERT book. Use with the separate revision notes.

Constants

h	6.63×10^{-34} J s	c	3×10^{8} m s^{-1}
e	1.6×10^{-19} C	m_e	9.1×10^{-31} kg
m_p	1.673×10^{-27} kg	m_n	1.675×10^{-27} kg
1 u	1.66×10^{-27} kg $= 931.5$ MeV/c^2	N_A	6.023×10^{23} mol^{-1}
hc	1240 eV nm	R (Rydberg)	1.097×10^{7} m^{-1}
a_0 (Bohr radius)	0.53 Å	R_0 (nuclear)	1.2 fm
1 eV	1.6×10^{-19} J	1 MeV	1.6×10^{-13} J

Chapter 9 Ray Optics and Optical Instruments

Mirrors and refraction

Sign convention: distances measured from the pole/optical centre; along incident light +, against −; up +.

Mirror: $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$, $f=\frac{R}{2}$, $m=-\frac{v}{u}=\frac{h'}{h}$ (concave $f<0$, convex $f>0$)

Snell: $n_1 \sin i = n_2 \sin r$ $n=\frac{c}{v}$ apparent depth $=\frac{\text{real depth}}{n}$ slab shift $= t\left(1-\frac{1}{n}\right)$

Critical angle: $\sin i_c = \frac{1}{n}$ (denser to rarer, $n = n_{dense}/n_{rare}$) Spherical surface: $\frac{n_2}{v}-\frac{n_1}{u}=\frac{n_2-n_1}{R}$

Lenses and prism

Lens: $\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$, $m=\frac{v}{u}$ Lens maker: $\frac{1}{f}=(n_{21}-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$

Power $P=\frac{1}{f\ (\text{m})}$ [dioptre, D] In contact: $\frac{1}{f}=\frac{1}{f_1}+\frac{1}{f_2}$, $P=P_1+P_2$, $m=m_1m_2$

Prism: $A=r_1+r_2$, $\delta=i+e-A$, $n=\frac{\sin\frac{A+D_m}{2}}{\sin\frac{A}{2}}$ thin prism $\delta=(n-1)A$

Optical instruments ($D = 25$ cm)

	Image at D	Image at ∞ (normal)
Simple microscope	$m = 1 + \dfrac{D}{f}$	$m = \dfrac{D}{f}$
Compound microscope	$m = \dfrac{v_o}{\lvert u_o\rvert}\left(1 + \dfrac{D}{f_e}\right) \approx \dfrac{L}{f_o}\left(1 + \dfrac{D}{f_e}\right)$	$m = \dfrac{L}{f_o} \cdot \dfrac{D}{f_e}$
Astronomical telescope	$m = \dfrac{f_o}{f_e}\left(1 + \dfrac{f_e}{D}\right)$	$m = \dfrac{f_o}{f_e}$, $L = f_o + f_e$

JEE extra

Displacement method: $f = \dfrac{D^2 - d^2}{4D}$, needs $D \geq 4f$. Lenses separated by d: $\dfrac{1}{F} = \dfrac{1}{f_1} + \dfrac{1}{f_2} - \dfrac{d}{f_1 f_2}$.
Silvered lens: $\dfrac{1}{F} = \dfrac{2}{f_l} + \dfrac{1}{f_m}$.
Mirror rotated by θ: reflected ray turns 2θ. Fibre: $\sin i_{max} = \sqrt{n_1^2 - n_2^2}$. Longitudinal magnification $\approx -m^2$. Dispersive power $\omega = \dfrac{n_v - n_r}{n_y - 1}$.

Chapter 10 Wave Optics

Huygens, interference, diffraction, polarisation

In a medium: ν unchanged, $v = \dfrac{c}{n}$, $\lambda = \dfrac{\lambda_0}{n}$ Snell (Huygens): $\dfrac{\sin i}{\sin r} = \dfrac{v_1}{v_2}$

Phase difference $\phi = \dfrac{2\pi}{\lambda} \times$ path difference $I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$; equal: $I = 4I_0 \cos^2\dfrac{\phi}{2}$

YDSE: bright $x_n = \dfrac{n\lambda D}{d}$; dark $x_n = \left(n + \frac{1}{2}\right)\dfrac{\lambda D}{d}$; fringe width $\beta = \dfrac{\lambda D}{d}$; angular width $\dfrac{\lambda}{d}$

Single slit (width a): minima $a\sin\theta = n\lambda$; first minimum $\theta = \dfrac{\lambda}{a}$; central maximum width $\dfrac{2\lambda D}{a}$

Malus: $I = I_0 \cos^2\theta$; unpolarised through a polaroid: $I = \dfrac{I_0}{2}$

JEE extra

$\dfrac{I_{max}}{I_{min}} = \left(\dfrac{\sqrt{I_1} + \sqrt{I_2}}{\sqrt{I_1} - \sqrt{I_2}}\right)^2$. Sheet over a slit: shift $= \dfrac{(n-1)tD}{d}$. Brewster: $\tan i_B = n$. Fresnel distance $z_F = \dfrac{a^2}{\lambda}$. Resolving power of telescope $\dfrac{D}{1.22\lambda}$.

Chapter 11 Dual Nature of Radiation and Matter

Photons and photoelectric effect

Photon: $E = h\nu = \dfrac{hc}{\lambda}$ [J]; $E(\text{eV}) = \dfrac{1240}{\lambda(\text{nm})}$; $p = \dfrac{h}{\lambda}$ [kg m s^{-1}]; photons per second $N = \dfrac{P}{h\nu}$

Einstein: $K_{max} = h\nu - \phi_0 = eV_0$ threshold $\nu_0 = \dfrac{\phi_0}{h}$, $\lambda_0 = \dfrac{hc}{\phi_0}$ $V_0 = \dfrac{h}{e}\nu - \dfrac{\phi_0}{e}$ (slope $\dfrac{h}{e}$)

X-ray cut-off from electrons through V: $\nu_{max} = \dfrac{eV}{h}$, $\lambda_{min} = \dfrac{hc}{eV}$

Matter waves

$\lambda = \dfrac{h}{p} = \dfrac{h}{mv} = \dfrac{h}{\sqrt{2mK}} = \dfrac{h}{\sqrt{2mqV}}$ electron: $\lambda = \dfrac{1.227}{\sqrt{V}}$ nm

JEE extra

Radiation force $F = \dfrac{P}{c}$ (absorbed), $\dfrac{2P}{c}$ (reflected). Thermal particle: $\lambda = \dfrac{h}{\sqrt{3mkT}}$. Same V: $\lambda \propto \dfrac{1}{\sqrt{qm}}$; same K: $\lambda \propto \dfrac{1}{\sqrt{m}}$.

Chapter 12 Atoms

Rutherford and Bohr

Closest approach: $d = \dfrac{1}{4\pi\varepsilon_0}\dfrac{2Ze^2}{K}$ Bohr: $mvr = \dfrac{nh}{2\pi}$, $h\nu = E_i - E_f$

$r_n = \dfrac{n^2h^2\varepsilon_0}{\pi me^2} = 0.53\,n^2$ Å $v_n = \dfrac{e^2}{2\varepsilon_0 nh} = \dfrac{2.18\times10^6}{n}$ m s^{-1} $E_n = -\dfrac{13.6}{n^2}$ eV

$K = -E$, $U = 2E$ Rydberg: $\dfrac{1}{\lambda} = R\left(\dfrac{1}{n_f^2} - \dfrac{1}{n_i^2}\right)$ lines from level n: $\dfrac{n(n-1)}{2}$

Series: Lyman $n_f = 1$ (UV); Balmer 2 (visible); Paschen 3, Brackett 4, Pfund 5 (IR) de Broglie: $2\pi r_n = n\lambda$

JEE extra

H-like ion: $r_n = \dfrac{0.53n^2}{Z}$ Å, $E_n = -\dfrac{13.6Z^2}{n^2}$ eV, $v_n \propto \dfrac{Z}{n}$, $T \propto \dfrac{n^3}{Z^2}$. Recoil speed $v = \dfrac{h\nu}{Mc}$. Impact parameter $b = \dfrac{Ze^2\cot(\theta/2)}{4\pi\varepsilon_0 K}$.

Chapter 13 Nuclei

Size, mass defect, binding energy

$R = R_0 A^{1/3}$ density $\approx 2.3 \times 10^{17}$ kg m^{-3} (independent of A)

$\Delta M = Z m_H + (A - Z) m_n - M_{atom}$ $E_b = \Delta M\, c^2 = \Delta M(\text{u}) \times 931.5$ MeV $E_{bn} = \dfrac{E_b}{A}$

$E = mc^2$ $Q = (m_{reactants} - m_{products})c^2$: $Q > 0$ exothermic

E_{bn}: max ≈ 8.75 MeV at $A = 56$; ≈ 7.6 MeV at $A \approx 238$. Fission ≈ 200 MeV per ^{235}U.

Isotopes: same Z; isobars: same A; isotones: same $N = A - Z$.

JEE extra

Decay: $N = N_0 e^{-\lambda t}$, $T_{1/2} = \dfrac{0.693}{\lambda}$, $\tau = \dfrac{1}{\lambda}$, $A = \lambda N$, $N = \dfrac{N_0}{2^{t/T_{1/2}}}$. α: $(A-4, Z-2)$; β^-: $Z+1$; β^+: $Z-1$. $K_\alpha = Q\dfrac{A-4}{A}$.

Chapter 14 Semiconductor Electronics

Semiconductors and diodes

Band gap: conductor (overlap), semiconductor ($E_g < 3$ eV: C 5.4, Si 1.1, Ge 0.7 eV), insulator ($E_g > 3$ eV)

Intrinsic: $n_e = n_h = n_i$ Extrinsic: $n_e n_h = n_i^2$; n-type $n_e \gg n_h$ (pentavalent); p-type $n_h \gg n_e$ (trivalent)

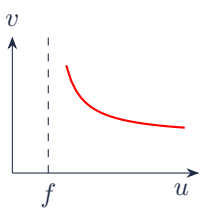
Barrier potential ≈ 0.7 V (Si) Forward bias: barrier lowered, depletion narrows; reverse: barrier raised, widens

Rectifier output frequency: half-wave $= f_{in}$; full-wave $= 2f_{in}$

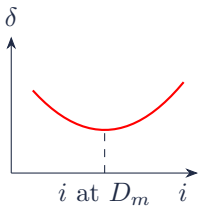
JEE extra

$\sigma = e(n_e\mu_e + n_h\mu_h)$. Diode: $I = I_0(e^{eV/kT} - 1)$. Dynamic resistance $r_d = \dfrac{\Delta V}{\Delta I}$. Gates: AND $A \cdot B$, OR $A + B$, NOT $\bar{A}$, NAND $\overline{AB}$, NOR $\overline{A + B}$ (NAND, NOR universal).

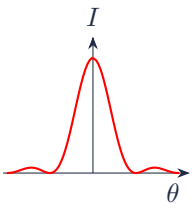
Graph shapes to remember



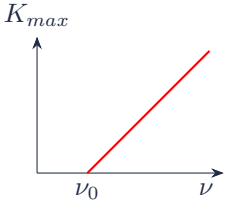
real image: $|v|$ vs $|u|$



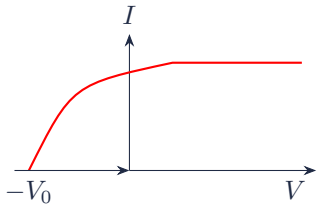
prism: δ vs i



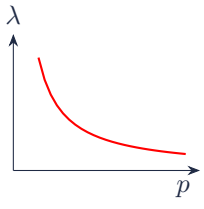
single slit: wide central max



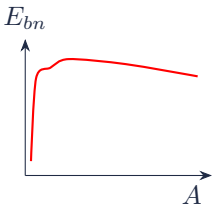
K_{max} vs ν (slope h)



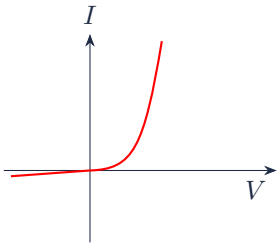
photocurrent vs V



de Broglie: $\lambda \propto 1/p$



BE/nucleon: peak at $A = 56$



diode: forward and reverse